SUMS OF POWERS OF INTEGERS: A COMPLETE FRAMEWORK FOR CLOSED FORMULAS

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Abstract.

Contents

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1. Definitions and notations

Allow us to define the mathematical objects we utilize throughout the manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], because it is simple and beautiful.

Definition 1.1 (Multifold sums of powers)**.** *For non-negative integers r, n, m*

$$\begin{aligned}\Sigma^0 n^m &= n^m,\\ \Sigma^1 n^m &= \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,\\ \Sigma^{r+1} n^m &= \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.\end{aligned}$$

We utilize the following notation for central factorials

Definition 1.2 (Central factorials)**.** *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0,\\ 1, & \text{if } k = 0,\\ n\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = n\left(n+\frac{k}{2}-1\right)_{k-1}, & \text{if } k > 0,\end{cases}$$

where $\left(n+\frac{k}{2}-1\right)_{k-1} = \prod_{j=1}^{k-1}\left(n+\frac{k}{2}-j\right)$ are falling factorials.

J. F. Steffensen gives quite a good discussion on factorial notations and definitions in [2].

2. Historical context

3. Helper lemmas

Lemma 3.1 (Central Newton's formula)**.** *For a function f defined on integers and arbitrary integers x, t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!}\delta^k f(t),$$

where $\delta^k f(t)$ denotes the k-th central finite difference

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} f\left(t + \tfrac{k}{2} - j\right),$$

and $(x-t)^{[k]}$ are central factorials defined by (1.2).

Proof. See [3, p. 462], and [4, section 2, eq. (19)], and [5, section 6, eq. (21)]. □

Lemma 3.2 (Newton's formula for powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Lemma 3.3 (Central factorials binomial form)**.** *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!}\left(n+\frac{k}{2}-1\right)_{k-1} = \frac{n}{k(k-1)!}\left(n+\frac{k}{2}-1\right)_{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

Lemma 3.4 (Central factorials binomial form decomposition)**.** *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\binom{n+\frac{k}{2}}{k} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k}\binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n+\frac{k}{2}}{k} = \frac{n+\frac{k}{2}}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

Thus,

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \binom{n+\frac{k}{2}}{k} - \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\left[2\binom{n+\frac{k}{2}}{k} - \binom{n+\frac{k}{2}-1}{k-1}\right] = \frac{1}{2}\left[\binom{n+\frac{k}{2}}{k} + \binom{n+\frac{k}{2}-1}{k} + \binom{n+\frac{k}{2}-1}{k-1} - \binom{n+\frac{k}{2}-1}{k-1}\right].$$

Thus, the statement follows. □

Lemma 3.5 (Newton's formula for powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k}\right]\delta^k t^m.$$

Lemma 3.6 (Generalized hockey-stick identity). *For integers a, b and j,*

$$\sum_{k=a}^{b}\binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^{b}\binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right)$. By hockey stick identity $\sum_{k=0}^{n}\binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b}\binom{k}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 3.7 (Centered hockey-stick identity). *For integers $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $j = 1-t+\frac{k}{2}+r$, and $b = n-t+\frac{k}{2}+r$ yields

$$\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r}\binom{j}{k}.$$

By generalized hockey-stick identity (3.6),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r}\binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows. □

4. Ordinary sums of powers

Proposition 4.1 (Ordinary sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^{m} \delta^k t^m \sum_{j=1}^{n} \left[\binom{j - t + \frac{k}{2}}{k} + \binom{j - t + \frac{k}{2} - 1}{k} \right].$$

Proposition 4.2 (Ordinary sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^{m} \left[\binom{n - t + \frac{k}{2} + 1}{k+1} - \binom{1 - t + \frac{k}{2}}{k+1} + \binom{n - t + \frac{k}{2}}{k+1} - \binom{-t + \frac{k}{2}}{k+1} \right] \delta^k t^m.$$

For example,

Example 4.3 (Sums of squares in central differences)**.**

$$t = 0: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(1+n) + \frac{\delta^2 t^2}{2}\, \frac{n(1 + 3n + 2n^2)}{6},$$

$$t = 1: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-1) + \frac{\delta^2 t^2}{2}\, \frac{n(1 - 3n + 2n^2)}{6},$$

$$t = 2: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-3) + \frac{\delta^2 t^2}{2}\, \frac{n(13 - 9n + 2n^2)}{6},$$

$$t = 3: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-5) + \frac{\delta^2 t^2}{2}\, \frac{n(37 - 15n + 2n^2)}{6},$$

$$t = 4: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-7) + \frac{\delta^2 t^2}{2}\, \frac{n(73 - 21n + 2n^2)}{6}.$$

Example 4.4 (Sums of cubes in central differences)**.**

$$t=0: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(1+n) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(1+3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-1+n+4n^2+2n^3)}{24},$$

$$t=1: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-1) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(1-3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(1+n-4n^2+2n^3)}{24},$$

$$t=2: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-3) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(13-9n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-21+25n-12n^2+2n^3)}{24},$$

$$t=3: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-5) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(37-15n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-115+73n-20n^2+2n^3)}{24},$$

$$t=4: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-7) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(73-21n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-329+145n-28n^2+2n^3)}{24}.$$

It is indeed interesting to notice the connection between the decomposition of ordinary sums of powers (4.2) and Riordan's central factorial numbers of the second kind, that is

Proposition 4.5 (Central factorial numbers sums)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n+\frac{k}{2}+1}{k+1} - \binom{1+\frac{k}{2}}{k+1} + \binom{n+\frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1}\right] k!\,T(m,k),$$

where $T(n,k)$ denotes the central factorial numbers of the second kind (in Riordan's notation). See [5, p. 213]*, and* [6].

These central factorial numbers of the second kind $T(n,k)$ are defined by central Newton's formula for powers (3.2) evaluated in zero, such that

$$n^m = \sum_{k=0}^{\infty} T(m,k) n^{[k]}.$$

Which implies that

$$T(n,k) = \frac{\delta^k 0^m}{k!},$$

by Newton's formula (3.2).

5. Remark on central factorial numbers of the second kind

6. Generalized central factorial numbers of the second kind

Definition 6.1 (Generalized central factorial numbers of the second kind)**.** *For integers $0 \le k \le m$, and an arbitrary integer t, the generalized central factorial numbers of the second kind are defined by the relation*

$$n^m = \sum_{k=0}^{\infty} T_t(m,k) \cdot (n-t)^{[k]}.$$

It is also straightforward that

$$T_t(m,k) = \frac{\delta^k t^m}{k!},$$

by Newton's formula (3.2). For example,

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	0	1							
2	0	0	1						
3	0	$\frac{1}{4}$	0	1					
4	0	0	1	0	1				
5	0	$\frac{1}{16}$	0	$\frac{5}{2}$	0	1			
6	0	0	1	0	5	0	1		
7	0	$\frac{1}{64}$	0	$\frac{91}{16}$	0	$\frac{35}{4}$	0	1	
8	0	0	1	0	21	0	14	0	1

Table 1. Values of generalized central factorial numbers $T_0(n,k)$. See also the sequences A000000, and A000000 in [7].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	1	1							
2	1	2	1						
3	1	$\frac{13}{4}$	3	1					
4	1	5	7	4	1				
5	1	$\frac{121}{16}$	15	$\frac{25}{2}$	5	1			
6	1	$\frac{91}{8}$	31	35	20	6	1		
7	1	$\frac{1093}{64}$	63	$\frac{1491}{16}$	70	$\frac{119}{4}$	7	1	
8	1	$\frac{205}{8}$	127	$\frac{483}{2}$	231	126	42	8	1

Table 2. Values of generalized central factorial numbers $T_1(n, k)$. See also the sequences A000000, and A000000 in [7].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	2	1							
2	4	4	1						
3	8	$\frac{49}{4}$	6	1					
4	16	34	25	8	1				
5	32	$\frac{1441}{16}$	90	$\frac{85}{2}$	10	1			
6	64	$\frac{931}{4}$	301	190	65	12	1		
7	128	$\frac{37969}{64}$	966	$\frac{12411}{16}$	350	$\frac{371}{4}$	14	1	
8	256	$\frac{6001}{4}$	3025	3003	1701	588	126	16	1

Table 3. Values of generalized central factorial numbers $T_2(n, k)$. See also the sequences A000000, and A000000 in [7].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	3	1							
2	9	6	1						
3	27	$\frac{109}{4}$	9	1					
4	81	111	55	12	1				
5	243	$\frac{6841}{16}$	285	$\frac{185}{2}$	15	1			
6	729	$\frac{12753}{8}$	1351	585	140	18	1		
7	2187	$\frac{372709}{64}$	6069	$\frac{53011}{16}$	1050	$\frac{791}{4}$	21	1	
8	6561	$\frac{167943}{8}$	26335	$\frac{35049}{2}$	6951	1722	266	24	1

Table 4. Values of generalized central factorial numbers $T_3(n,k)$. See also the sequences A000000, and A000000 in [7].

t	$T_t(2m, 2k)$	OEIS ID
0	$1, 0, 1, 0, 1, 1, 0, 1, 5, 1, 0, 1, 21, 14, 1, 0, 1, 85, 147, 30, 1, \cdots$	A269945
1	$1, 1, 1, 1, 7, 1, 1, 31, 20, 1, 1, 127, 231, 42, 1, 1, 511, \cdots$	A394692
2	$1, 4, 1, 16, 25, 1, 64, 301, 65, 1, 256, 3025, 1701, 126, 1, \cdots$	A395456
3	$1, 9, 1, 81, 55, 1, 729, 1351, 140, 1, 6561, 26335, 6951, 266, 1, \cdots$	A395457

Table 5. Sequences of generalized central factorial numbers $T_t(2m, 2k)$.

Thus, we have the formula for sums of powers in terms of generalized central factorial numbers of the second kind.

Proposition 6.2 (Generalized Central factorial sums)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1}\right] k! T_t(m,k).$$

7. Double and triple sums of powers in terms of central differences

Proposition 7.1 (Double sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^2 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[+\binom{n-t+\frac{k}{2}+2}{k+2}-\binom{2-t+\frac{k}{2}}{k+2}\Sigma^0 n^0-\binom{1-t+\frac{k}{2}}{k+1}\Sigma^1 n^0$$
$$+\binom{n-t+\frac{k}{2}+1}{k+2}-\binom{1-t+\frac{k}{2}}{k+2}\Sigma^0 n^0-\binom{0-t+\frac{k}{2}}{k+1}\Sigma^1 n^0\Bigg]\delta^k t^m$$

For example,

Example 7.2 (Double sum of squares in central differences).

$$t=0: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(1+n)^2(2+n)}{12},$$

$$t=1: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-1+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n^2(-1+n^2)}{12},$$

$$t=2: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-4-3n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(10+5n-4n^2+n^3)}{12},$$

$$t=3: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-7-6n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(32+23n-8n^2+n^3)}{12},$$

$$t=4: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-10-9n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(66+53n-12n^2+n^3)}{12}.$$

Proposition 7.3 (Triple sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^3 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[$$
$$+\binom{n-t+\frac{k}{2}+3}{k+3}-\binom{3-t+\frac{k}{2}}{k+3}\Sigma^0 n^0-\binom{2-t+\frac{k}{2}}{k+2}\Sigma^1 n^0-\binom{1-t+\frac{k}{2}}{k+1}\Sigma^2 n^0$$
$$+\binom{n-t+\frac{k}{2}+2}{k+3}-\binom{2-t+\frac{k}{2}}{k+3}\Sigma^0 n^0-\binom{1-t+\frac{k}{2}}{k+2}\Sigma^1 n^0-\binom{0-t+\frac{k}{2}}{k+1}\Sigma^2 n^0\Bigg]\delta^k t^m$$

For example,

Example 7.4 (Triple sum of squares in central differences)**.**

$$t=0: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(6+11n+6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(18+45n+40n^2+15n^3+2n^4)}{120},$$

$$t=1: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-2-n+2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(-2-5n+5n^3+2n^4)}{120},$$

$$t=2: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-10-13n-2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(58+65n-5n^3+2n^4)}{120},$$

$$t=3: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-18-25n-6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(198+255n+40n^2-15n^3+2n^4)}{120},$$

$$t=4: \quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-26-37n-10n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(418+565n+120n^2-25n^3+2n^4)}{120}.$$

8. Multifold sums of powers in terms of central differences

Proposition 8.1 (Multifold sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\begin{aligned}\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{&\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r}\\ &-\sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\Bigg\}\delta^k t^m.\end{aligned}$$

For example,

Example 8.2 (Examples of multifold sums)**.**

$$
\begin{aligned}
t=0: \quad \Sigma^r n^m &= \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n+\frac{k}{2}+r}{k+r}+\tbinom{n+\frac{k}{2}+r-1}{k+r} \\
&\qquad -\sum_{s=0}^{r-1}\Big[\tbinom{r-s+\frac{k}{2}}{k+r-s}+\tbinom{r-1-s+\frac{k}{2}}{k+r-s}\Big]\Sigma^s n^0\Bigg\}\delta^k 0^m, \\
t=1: \quad \Sigma^r n^m &= \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n-1+\frac{k}{2}+r}{k+r}+\tbinom{n-1+\frac{k}{2}+r-1}{k+r} \\
&\qquad -\sum_{s=0}^{r-1}\Big[\tbinom{r-s-1+\frac{k}{2}}{k+r-s}+\tbinom{r-2-s+\frac{k}{2}}{k+r-s}\Big]\Sigma^s n^0\Bigg\}\delta^k 1^m, \\
t=2: \quad \Sigma^r n^m &= \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n-2+\frac{k}{2}+r}{k+r}+\tbinom{n-2+\frac{k}{2}+r-1}{k+r} \\
&\qquad -\sum_{s=0}^{r-1}\Big[\tbinom{r-s-2+\frac{k}{2}}{k+r-s}+\tbinom{r-3-s+\frac{k}{2}}{k+r-s}\Big]\Sigma^s n^0\Bigg\}\delta^k 2^m.
\end{aligned}
$$

9. Multifold sums of powers in generalized central factorial numbers

Proposition 9.1 (Multifold central factorial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$
\begin{aligned}
\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{&\binom{n-t+\frac{k}{2}+r}{k+r}+\binom{n-t+\frac{k}{2}+r-1}{k+r} \\
&-\sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s}+\binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\Bigg\}k!T_t(m,k).
\end{aligned}
$$

Example 9.2 (Sums of squares in central factorial numbers)**.**

$$
\begin{aligned}
t=0: &\quad \Sigma^1 n^2 = n\cdot 0 + \tfrac{1}{2}n(1+n)\cdot 0 + \tfrac{1}{6}n(1+3n+2n^2)\cdot 1,\\
t=1: &\quad \Sigma^1 n^2 = n\cdot 1 + \tfrac{1}{2}n(n-1)\cdot 2 + \tfrac{1}{6}n(1-3n+2n^2)\cdot 1,\\
t=2: &\quad \Sigma^1 n^2 = n\cdot 4 + \tfrac{1}{2}n(n-3)\cdot 4 + \tfrac{1}{6}n(13-9n+2n^2)\cdot 1,\\
t=3: &\quad \Sigma^1 n^2 = n\cdot 9 + \tfrac{1}{2}n(n-5)\cdot 6 + \tfrac{1}{6}n(37-15n+2n^2)\cdot 1,\\
t=4: &\quad \Sigma^1 n^2 = n\cdot 16 + \tfrac{1}{2}n(n-7)\cdot 8 + \tfrac{1}{6}n(73-21n+2n^2)\cdot 1,\\
t=5: &\quad \Sigma^1 n^2 = n\cdot 25 + \tfrac{1}{2}n(n-9)\cdot 10 + \tfrac{1}{6}n(121-27n+2n^2)\cdot 1.
\end{aligned}
$$

We may notice that

- For $t=0$ the coefficients ... correspond to the second row of the triangle [reference].
- For $t=1$ the coefficients ... correspond to the second row of the triangle [reference].
- For $t=2$ the coefficients ... correspond to the second row of the triangle [reference].
- For $t=3$ the coefficients ... correspond to the second row of the triangle [reference].

Example 9.3 (Sums of cubes in central factorial numbers)**.**

$$t=0: \quad \Sigma^1 n^4 = n\cdot 0 + \tfrac{1}{2}n(1+n)\cdot 0 + \tfrac{1}{6}n(1+3n+2n^2)\cdot 1$$
$$+ \tfrac{1}{8}n(-1+n+4n^2+2n^3)\cdot 0 + \tfrac{1}{10}n(-2-5n+5n^3+2n^4)\cdot 1,$$

$$t=1: \quad \Sigma^1 n^4 = n\cdot 1 + \tfrac{1}{2}n(n-1)\cdot 5 + \tfrac{1}{6}n(1-3n+2n^2)\cdot 7$$
$$+ \tfrac{1}{8}n(1+n-4n^2+2n^3)\cdot 4 + \tfrac{1}{10}n(-2+5n-5n^3+2n^4)\cdot 1,$$

$$t=2: \quad \Sigma^1 n^4 = n\cdot 16 + \tfrac{1}{2}n(n-3)\cdot 34 + \tfrac{1}{6}n(13-9n+2n^2)\cdot 25$$
$$+ \tfrac{1}{8}n(-21+25n-12n^2+2n^3)\cdot 8$$
$$+ \tfrac{1}{10}n(18-45n+40n^2-15n^3+2n^4)\cdot 1,$$

$$t=3: \quad \Sigma^1 n^4 = n\cdot 81 + \tfrac{1}{2}n(n-5)\cdot 111 + \tfrac{1}{6}n(37-15n+2n^2)\cdot 55$$
$$+ \tfrac{1}{8}n(-115+73n-20n^2+2n^3)\cdot 12$$
$$+ \tfrac{1}{10}n(298-275n+120n^2-25n^3+2n^4)\cdot 1$$

We may notice that

- For $t=0$ the coefficients ... correspond to the fourth row of the triangle [reference].
- For $t=1$ the coefficients ... correspond to the fourth row of the triangle [reference].
- For $t=2$ the coefficients ... correspond to the fourth row of the triangle [reference].
- For $t=3$ the coefficients ... correspond to the fourth row of the triangle [reference].

10. Multifold sums of powers in binomial form

Lemma 10.1 (Multifold sum of zero powers)**.** *For integers $r \geq 0$ and $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

Proof. (1) Let $r=0$, then we have $\Sigma^0 n^0 = 1$, by definition (1.1).

(2) Let $r=1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.

(3) Let $r=2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r=3$, then we have $\Sigma^3\, n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary integer r, by hockey stick identity $\sum_{k=1}^{n} \binom{k}{r} = \binom{n+1}{r+1}$, yields $\Sigma^r\, n^0 = \sum_{k=1}^{n} \Sigma^{r-1}\, k^0 = \sum_{k=1}^{n} \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

$\square$

Proposition 10.2 (Multifold binomial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r\, n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\Bigg\}\delta^k t^m$$

For example,

Example 10.3 (Examples of multifold binomial sums)**.**

$$t=0: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\Bigg\}\delta^k 0^m,$$

$$t=1: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\Bigg\}\delta^k 1^m,$$

$$t=2: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\Bigg\}\delta^k 2^m.$$

11. Related works

12. Framework for sums of powers

13. Conclusions

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14. Mathematica programs

Use this GitHub repository to validate the results using Mathematica programs.

Mathematica Function	Validates / Prints
`MultifoldSumOfPowersRecurrence[r, n, m]`	Computes $\Sigma^r n^m$, (1.1)
`CentralFactorial[n, k]`	Computes $n^{[k]}$, (1.2)
`CentralFactorialsBinomialFormDecomposition[20]`	Validates Lem. (3.4)
`ValidateNewtonsFormulaForPowers[10]`	Validates Lem. (3.5)
`ValidateCenteredHockeyStickIdentity[5]`	Validates Lem. (3.7)
`ValidateOrdinarySumsOfPowersDecomposition[10]`	Validates Prop. (4.2)
`ValidateDoubleSumsOfPowersDecomposition[5]`	Validates Prop. (7.1)
`ValidateTripleSumsOfPowersDecomposition[5]`	Validates Prop. (7.3)
`ValidateMultifoldSumsOfPowers[3]`	Validates Prop. (8.1)